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A Limitations

The limitations of our work are summarized as follows:

Monolingual Capability. ComplexBench is primarily constructed based on Chinese reference instructions, which may neglect some elements in other languages and cultures that can influence the complexity of instructions. Recognizing this constraint, we plan to expand ComplexBench by incorporating multiple languages to investigate the disparities in complex instruction-following ability of LLMs across different linguistic environments in future iterations.

LLM-based Evaluation. The evaluation method based on LLM is widely used in the automatic evaluation process of ComplexBench. Although experiments show that our evaluation method achieves satisfactory agreement with human judgment generally, the potential biases of LLM-as-Judge, such as verbosity and self-enhancement [5], may affect the overall evaluation correctness. Additionally, we utilize GPT-4-1106 commercial APIs for evaluation, which presents challenges such as high costs and potential data leakage. We leave the development of more accurate and efficient methods for evaluating complex instruction-following as important future work.

B Author Statement and License

ComplexBench is distributed under CC BY 4.0. The evaluation code of ComplexBench is distributed under the MIT license. We will bear all responsibility in case of violation of rights, etc.

C Task Distribution of ComplexBench

We refer to the taxonomy of AlignBench [6] to categorize the task types of instructions in the ComplexBench. Taking into account that instructions about mathematics have relatively fixed answers and are difficult to construct complex instructions, as well as the coarse granularity of the writing ability category. We remove mathematical and use 4 subcategories of writing ability in AlignBench: practical writing, creative writing, professional writing, and custom writing. When annotators construct instructions, they also provide task category labels simultaneously, the results are shown in Table 8.

Category	#Samples
Fundamental Language Ability	159
Advanced Chinese Understanding	62
Open-ended Questions	115
Practical Writing	195
Creative Writing	105
Professional Writing	183
Custom Writing	73
Logical Reasoning	107
Task-oriented Role Play	95
Professional Knowledge	56
Total	1150

Table 8: Task distribution of ComplexBench dataset.

D Details of Constraint Dimensions

D.1 Lexical Constraint

1. **Word Matching.** The response should accurately find the corresponding content of certain keywords in the given instruction.
2. **Keywords.** The response should (not) include certain keywords, or include several words from a keyword list.

D.2 Format Constraint

1. **JSON Format.** The entire response should be wrapped in JSON format.

2. **Markdown Format.** The response should follow specific Markdown formats, such as equations, headings, and tables.
3. **Bullets Format.** The response should (not) contain bullet points.
4. **Length.** Control the length of the response, including the number of words, sentences, paragraphs, etc. This constraint can be used in combination with others, such as controlling the number of bullet points or the number of keywords included.
5. **Start with.** Control the content at the beginning of the response.
6. **End with.** Control the content at the end of the response.
7. **Punctuation.** Control the punctuation that appears in the response.
8. **Template.** The response should mimic the format of the given output template.

D.3 Semantic Constraint

1. **Language Style.** The response should adhere to a specific language style. We use the taxonomy of CharacterGLM [10], which defines language style from multiple aspects such as formality, imitation of celebrities, context-specific scenes, and discourse features (like using style from a certain website, emoji, etc.).
2. **Personalization.** The response should align with certain character attributes.
3. **Topic.** The response should focus on a specific topic.
4. **Sentiment.** The response should contain specific emotions. We refer to the six fine-grained categories of ECM [38] for sentiment, named as *Like*, *Happy*, *Sad*, *Disgust*, *Angry*, *Other*.

D.4 Utility Constraint

1. **Helpfulness.** The response should follow task descriptions.
2. **Target Language.** The response should be in a specific language, such as simplified Chinese, traditional Chinese or English.
3. **Supportiveness.** The response should be faithful to input texts, answering based on the information provided in the text completely.
4. **Consistency.** The content of the response should be consistent and free of contradictions.
5. **Factuality.** The response should correspond with facts, which primarily applies to instructions with definitive answers such as mathematical and logical reasoning.

E Prompts in Rule-Augmented LLM-based Evaluation

E.1 Prompts for Extractor in Rule-Augmented LLM-based Evaluation

Table 9 provides the prompt template we used for the LLM extractor in Rule-Augmented LLM-based evaluation. And Table 10 provides an example of scoring object extraction and their English translation. To improve performance, we use 6 manually constructed in-context examples in the prompt. Considering that the extraction of content differs significantly when there are multiple scoring objects (e.g., scoring question *"Does each shot's dialogue in the model output start with an interrogative sentence?"*), compared to when there is only one scoring object (e.g., scoring question *"Does the title of the speech given by the model have no more than 10 characters?"*). We use different sets of in-context examples for these two situations.

E.2 Prompts for Evaluator in Rule-Augmented LLM-based Evaluation

Table 11 provides the prompt template we used for the LLM evaluator in Rule-Augmented LLM-based evaluation. And Table 12 provides an example of automatic evaluation and their English translation. We have also explored different settings where all scoring questions from the instruction are presented to the evaluation model simultaneously, or asking the evaluation model to choose "YES" or "NO" without analysis. Ultimately, we found that the current settings achieve the highest level of agreement with humans.